

Efficient Impact Analysis of Changes in the Requirements of Manufacturing Automation Systems

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Abstract: Various drivers, especially increasing market dynamics, force industrial enterprises to evolve their production facilities ever more frequently. Analyzing the impact of changed requirements on the production facilities early in the engineering process would facilitate the estimation of resulting costs and therefore reduce project risk. In industrial practice, requirements specifications are typically textual lists and only structured by a table of contents; an analysis of the impact of changes within the requirements specification is extensive, as no documented information is available about how the requirements and other elements of the requirements specification depend on each other.

To identify impacts of changes in a requirements specification more efficiently, this paper proposes a model-based approach that classifies the elements of a requirements specification according to their contents and based on the classification preselects automatically candidate elements of the requirements specification that are possibly affected by the change. This paper shows the application of the approach on the manufacturing automation systems domain by means of a pick and place unit (PPU) demonstrator. The technical validation demonstrates that in the example of the PPU the impact analysis is more efficient with a content model than on an unstructured list; the effort for impact analysis is reduced further when adapting the content model to the manufacturing automation systems domain.

Keywords: Model-based Requirements Engineering, Requirements Analysis, Impact Analysis, Manufacturing System, Knowledge Acquisition, Systems Engineering

1. INTRODUCTION

Industrial plants typically operate for several decades (Birkhofer et al., 2012). During this long life cycle, multiple changes in the requirements on the plant arise in order to ensure competitiveness of industrial enterprises. Addressing changes in automation systems is challenging due to the multiple disciplines involved, for example mechanics, electrics/electronics and software, that have varying evolution cycles which have to be considered (Li et al., 2012). In this context, estimating the effort of a change is essential towards project success. This challenge can be addressed by an *impact analysis* for “identifying what to modify to accomplish a change or identifying the possible consequences of a change” (Arnold and Bohner, 1996, p. 34). An impact analysis consists typically of two main tasks (IEEE Computer Society, 2014, p. 108): Firstly, the *identification* of all system and software products that a change request affects. Secondly, the development of an *estimate* about the resources which are needed to realize the change. Management of changes to manufacturing automation system designs is a relatively young research

field (Hamraz et al., 2013) and it is still an open challenge (Vogel-Heuser et al., 2014b).

Requirements are the input for the design of the developed system; furthermore alternative solutions can be evaluated against the requirements (Lo and Couturier, 2012). Providing the capability to understand the effect of a change as early as possible within the engineering process will increase the chance of project success. The challenge is here that in industrial practice, requirement specifications are “currently coined by [...] informal documents and artifacts (i.e. typically written text)” (Insaurralde and Zoitl, 2013), often given as a list of requirements and other elements. Informal documents give no rigorous information on how the contained information depends on each other. Consequently, the whole requirements specification would have to be examined to identify the parts of it that are affected by a change. This paper proposes an approach to reduce the amount of information to be examined. The approach uses a model to classify the requirements and other elements of the requirements specification according to their contents, therefore called content model for requirements engineering (in the remainder of the paper referred to as requirements model). The requirements model not only describes possible contents in a requirements specification, but furthermore indicates how these contents depend on

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Ver.	Op.	ID	Title	Capability
1	•	1	Transport	TRC requires a recycling working cell for automatically transporting tanks between two semiautomatic working positions.
	•	2	Metallic tank	TRC requires the recycling working cell to handle metallic tanks (containing hazardous material).
	•	3	Parallel handling	TRC requires the recycling working cell to handle at least three tanks in parallel.
	•	4	Operating scenario	The operator starts up the PPU. In automatic mode, the PPU shall automatically transports metallic tanks between two semiautomatic working positions (from the initial working position A to their final working position B). The PPU stops processing if no further tank is available at position A.
	•	5	Automatic mode	The PPU shall have an automatic mode.
	•	6	Transport (pos. A to B)	The PPU shall transport metallic tanks automatically from position A to position B.
	•	7	Stop transporting	The PPU shall stop transporting metallic tanks if no further tank is available at position A.
	•	8	Three tanks at pos. A	The PPU shall handle at least three metallic tanks at working position A.
	•	9	Three tanks at pos. B	The PPU shall handle at least three metallic tanks at working position B.
2	+	10	Label	TRC requires the recycling working cell to label hazardous material for traceability reasons.
	+	11	Plastic tank	TRC requires the recycling working cell to handle black plastic tanks (not containing hazardous material).
	Δ	4'	Operating scenario	The operator starts up the PPU. In automatic mode, the PPU shall automatically label metallic tanks. In automatic mode, the PPU shall automatically transports black plastic and metallic tanks between two semiautomatic working positions (from the initial working position A to their final working position B). The PPU stops processing if no further tank is available at position A.
	Δ	6'	Transport (pos. A to B)	The PPU shall transport black plastic and metallic tanks automatically from position A to position B.
	Δ	7'	Stop	The PPU shall stop transporting black plastic and metallic tanks if no further tank is available at position A.
	+	12	Label metallic tanks	The PPU shall automatically label metallic tanks.

Fig. 1. Requirement specification of the PPU for both versions (Ver.) including modification operations (Op.) • (none), + (add), Δ (change).

each other. These dependencies indicate which contents might be affected by a change. Based on the requirements model, a subset of the requirements specification can be preselected automatically, consisting of possible candidates to be affected by a change. As not the whole requirements specification, but only preselected subsets and therefore less information have to be examined, the effort for an impact analysis is reduced. The smaller the preselected subsets are, the less information has to be examined, leading to an additional increase of efficiency. We demonstrate this gain of efficiency by a domain-specific tailoring of the approach.

The paper is structured as follows: Related work of the approach is discussed in Section 2. In Section 3, the pick and place unit case study is introduced which is used as a running example throughout this paper as well as for evaluation. The approach is presented in Section 4. The technical evaluation of the approach is presented in Section 5. The paper is concluded in Section 6.

2. RELATED WORK

The Association of German Engineers (VDI, 2014) provides a guideline for the sections and their contents that should be documented in a requirements specification in manufacturing automation systems. Other approaches classify requirements with the purpose of verification (Ji et al., 2011) and validation (Hackenberg et al., 2014) of requirements, in the latter case for the domain of mechatronic systems. We could not identify approaches in the manufacturing automation systems domain that specifically developed models such as requirements models to classify requirements in order to support impact analysis.

In the domain of software and systems engineering, modeling languages used in requirements engineering (Sikora et al., 2012) such as SysML (OMG, 2012) and UML (OMG, 2011) do not offer detailed models for the classification of requirements, but offer several diagrams to model a wide range of contents. In SysML requirements are explicitly addressed; requirements are visualized in diagrams and requirements tables.

Some approaches investigate change impact analyses for engineering models, even interdisciplinary ones. For example, matrix-based approaches are applied for change prediction in rotorcraft design (Clarkson et al., 2004) and for identifying high change risks in system models (Lindemann et al., 2009). Goering and Fay (2013) provide a meta-model for the domain of instrumentation and control systems and propose methods for analyzing the respective system designs.

Most approaches in industrial automation tackle evolution on code level (Briand et al., 2003; Vyatkin, 2013). Briand et al. (2003) and Müller and Rumpe (2014) developed approaches where explicit impact rules can be specified in a domain specific language and then be applied to UML models.

However, none of the above approaches addresses impact analysis on the requirements level. Approaches for impact analysis similar to the one presented in this paper have been developed in software and systems engineering. These approaches explicitly document traces based on models (Goknil et al., 2011; Mader et al., 2009; von Knethen, 2002; Ramesh and Jarke, 2001). The effort to document traces has to be compared to the resulting benefits. In contrast, the approach presented in this paper does not demand to

document traces explicitly as they are derived from the requirements model.

In summary, various approaches for requirements specification and impact analysis already exist in different domains. Nevertheless, maintaining and evolving a manufacturing systems engineering model or even analyzing the impact of change remains an open challenge (Vogel-Heuser et al., 2014b).

3. RUNNING EXAMPLE

For demonstration and technical validation of the proposed approach, an automated manufacturing system demonstrator is applied in the remainder of this paper. It is referred to as pick and place unit (PPU) henceforth. Various well-documented versions of the PPU are defined which can be combined in order to investigate different aspects of system evolution (Institute of Automation and Information Systems, Technische Universität München, 2014). As described by Legat et al. (2013), the PPU can be understood to be an evolving tank recycling machine operated by a fictitious tank recycling company (TRC). Two successive versions of the PPU are presented in the following. The requirements specification for both the initial and the evolved version are given in Fig. 1 as a list of requirements. An overview of the PPU’s mechanical layout in both versions applied here is provided in Fig. 2.

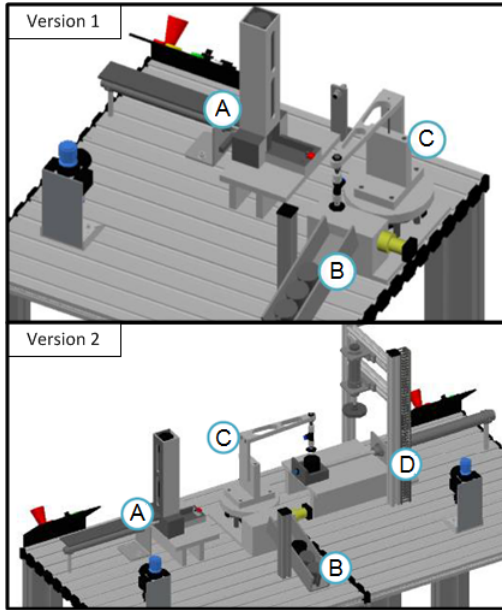


Fig. 2. Mechanical layout of two versions of the pick and place unit (Vogel-Heuser et al., 2014a).

Initially in Version 1, the PPU is a recycling machine automatically transporting tanks between two semi-automatic working positions (referred to as positions A and B). It solely handles metallic tanks containing hazardous material. Tanks are initially stored in a stack (Fig. 2, #A) at position A, transported by a crane (#C), and finally stored in a ramp (#B) at position B. The PPU is required to handle at least three tanks in parallel. Therefore, the stack and the ramp are designed to accommodate three tanks. The requirements specification for the initial version is given in Fig. 1 and marked with Ver. 1.

After a while operating the PPU, it became necessary to also handle black tanks containing non-hazardous material. This inferred an additional requirement to trace metallic tanks containing hazardous material. These new requirements resulted in the need to modify other requirements. The modification of the requirements specification for the evolved version is marked with Ver. 2 in Fig. 1. The modification includes the requirements that are added because they are new, and the requirements that had to be changed or added in consequence of the new requirements. The modified requirements lead to the implementation of an additional label module (Fig. 2, #D).

4. PROPOSED APPROACH

(a) *Generic Requirements Model* The purpose of the proposed approach is to efficiently identify those requirements and other elements of a requirements specification for a manufacturing automation system, which potentially are influenced by some change. As illustrated in Fig. 3, we divide this process into three steps: (1) *Manual classification* of requirements according to the requirements model in use, (2) *automatic preselection* of candidates according to the classification, and (3) *manual selection* of the candidates according to the actual impacts. In the following we describe each step in detail.

4.1 Manual Classification

Manual classification is concerned with assigning each element of the requirements specification to a respective requirements model element. To simplify the explanations, we assume that the requirements specification is given as a list (e.g. see Fig. 1). The approach requires a requirements model describing the types of elements in a requirements specification and their dependencies. The performance of the approach depends on the suitability of the employed requirements model for the given domain. The investiga-

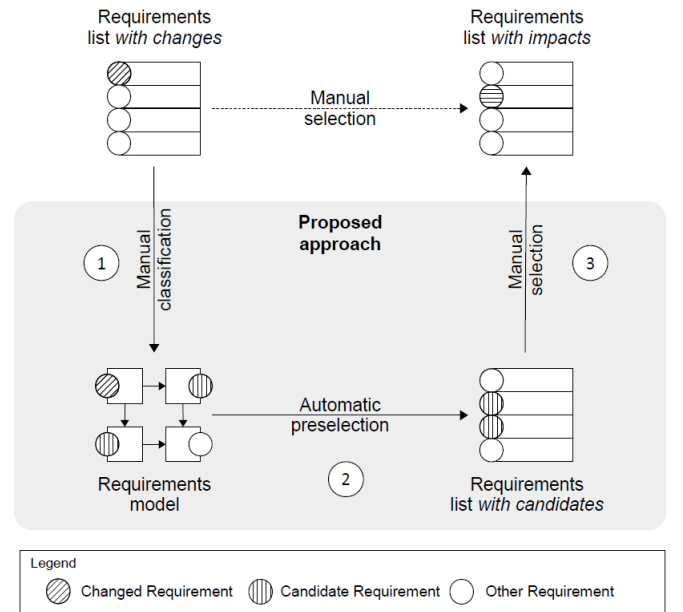
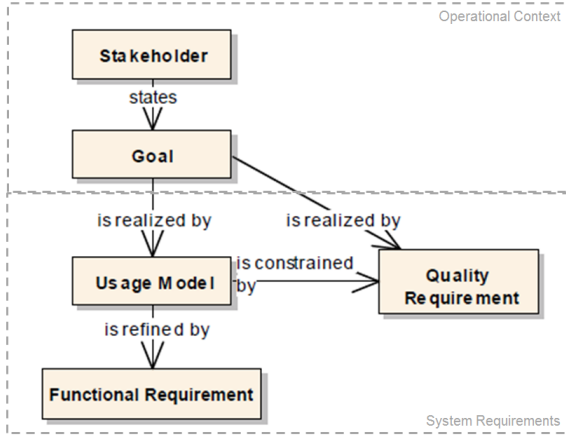
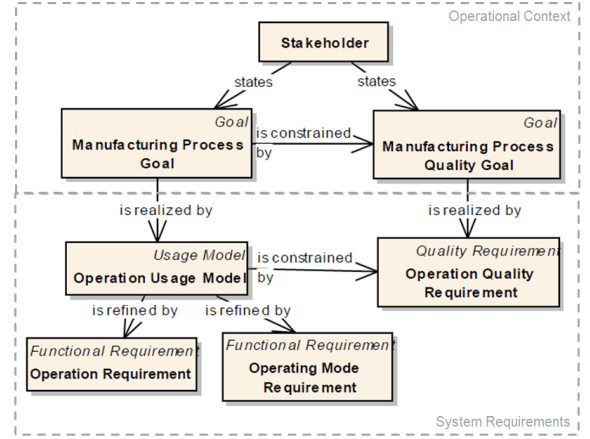


Fig. 3. Overview of the proposed approach with manual and automatic analysis steps.



(a) Generic requirements model (Penzenstadler and Eckhardt, 2012)



(b) Domain-specific requirements model

Fig. 4. The domain-specific requirements model is based on the generic requirements model and adds domain-specific characteristics.

tion of existing requirements models yielded that no dedicated model for the manufacturing automation systems domain exists. Therefore, we chose a *generic requirements model* for the cyber-physical systems domain (Penzenstadler and Eckhardt, 2012). Due to its scope on systems engineering and embedded systems it also addresses the manufacturing automation systems domain. To evaluate possible efficiency improvements, we further tailored the generic requirements model based on the requirements specification of the running example to obtain a *domain-specific requirements model*. In the following we describe both the generic and the domain-specific requirements model in detail.

An overview of the generic requirements model from (Penzenstadler and Eckhardt, 2012) is shown in Fig. 4a. In general, two types of requirements model elements can be distinguished: (1) The *operational context* and (2) *system requirements*. As indicated by the names, the first is concerned with the context of the system, while the second is concerned with the system itself.

The operational context is further divided into *stakeholders* and their *goals*. The list of stakeholders is usually determined to obtain all relevant persons and organizations, which possess some interest in the development project and/or the manufacturing automation system. For example, an important stakeholder of the PPU is the company TRC. Goals are "prescriptive statements of intent of one or more stakeholders" (van Lamsweerde, 2001). Besides others, goals typically include the original purpose of the system such as the transport of material between two semi-automatic working positions.

The system requirements are divided into *usage model*, *functional requirements* and *quality requirements*. The usage model is meant to realize stakeholder goals and describes observable system behavior including the interaction between users and the system. Note that according to Cockburn (2000) a usage model can be described by means of *use cases* and *scenarios*. The running example includes a scenario describing the interaction between operator, system and observed tank positions. Functional requirements

refine the usage model and represent units of functionality that are meant to realize the specified behavior. For example, the automatic mode requirement can be classified as a functional requirement. Quality requirements realize goals and constrain usage models by prescribing desired properties and conditions of the structure or behavior of the system. A sample property is the parallel handling of three tanks at position A and B.

(b) Domain-Specific Requirements Model We manually derive the domain-specific requirements model from the generic requirements model such that efficiency improvements through a better requirements model suitability can be demonstrated (see Sec. 5). An overview of the domain-specific requirements model is shown in Fig. 4b. The derived model keeps the distinction between (1) *operational context* and (2) *system requirements*.

The operational context again includes the *stakeholders*, while this time two types of goals are distinguished: The *manufacturing process goals* and the *manufacturing process quality goals*. Following the definition for manufacturing processes given by Kern (1979), manufacturing process goals encompass the transformation of physical and immaterial goods (i.e. material, energy, and information) required by a stakeholder. A sample manufacturing process goal is the transportation of tanks between two semi-automatic working positions. Manufacturing process quality goals prescribe the qualities of the manufacturing process such as required throughput or plant layout. The parallel handling of at least three tanks represents a typical manufacturing process quality goal.

The system requirements again include *(operation) usage models* and *(operation) quality requirements*, while this time two types of functional requirements are distinguished: The *operation requirement* and the *operation mode requirement*. Operation requirements refine operation usage models and represent operational units, which are used to accomplish the desired system behavior. One example is the transport of tanks from position A to B, which serves to accomplish the operation usage model and the manufacturing process goal respectively. Opera-

tion mode requirements prescribe necessary modes such as setting mode or automatic mode, which can be found in respective standards like DIN EN 13128 (DIN, 2009). For example the PPU includes an automatic mode for handling tanks without operator intervention.

4.2 Automatic Preselection

Automatic preselection is concerned with deriving a list of candidates from the requirements specification, which might be influenced by a change. The proposed preselection strategy assumes that the manual classification has been carried out and the assignment of the elements of the requirements specification to requirements model elements is available. In the following elements of the requirements specification are referred to as requirements; elements of the requirements model are referred to as elements. Fig. 5 illustrates the proposed strategy using an arbitrary requirements model consisting of three elements and nine requirements as well as the respective assignments. In the illustration, the first requirement assigned to element A is changed. Note that a change can be one of the operations *add*, *modify* or *delete*.

The preselection strategy adds all requirements assigned to the same model element (i.e. element A in our case) in the candidate list. All requirements assigned to subsequent neighbor elements (i.e. element B in our case) are also added to the candidate list. A neighbor element is subsequent if it has an incoming dependency from the first element, which is why element C and its assigned requirements are disregarded in the given example though being related to element A.

4.3 Manual Selection

Manual selection filters the candidate list returned by automatic preselection to contain only those requirements actually influenced by the change at hand. At this point we do not provide any specific support; rather the list of candidates has to be scanned manually to identify actual impacts. An example for the generic content model: After adding the plastic tank and labeling goals, only the usage models and quality requirements have to be processed

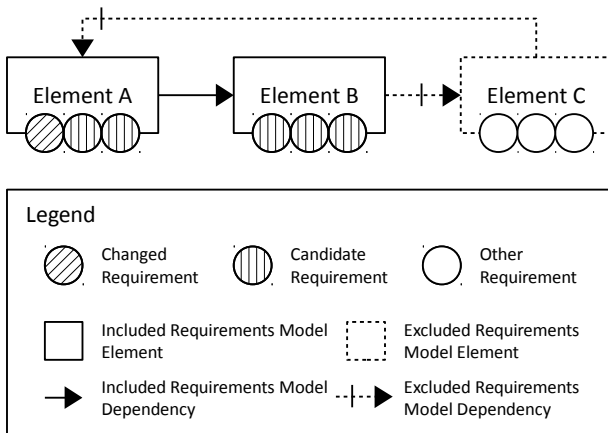


Fig. 5. Illustration of the preselection strategy based on the assignment of requirements to requirements model elements.

during manual selection according to the preselection strategy. The functional requirements can be omitted at this stage. Consequently a performance improvement is achieved compared to manual selection on the complete requirements list.

5. TECHNICAL VALIDATION

We provide a first technical validation of the proposed approach with respect to its *applicability* and *efficiency*. We first explain the validation method employed before carrying out the study and drawing conclusions from the results. Finally, we discuss the validity of our arguments with respect to the assumptions made.

5.1 Validation Method

At the current stage of our work we are interested in two main aspects of our approach, namely its applicability and its efficiency. By *applicability* we mean that the elements of the employed requirements models are suitable to describe the elements of the requirements specification for the running example. By *efficiency*, we mean that the proposed approach actually reduces the effort compared to impact analysis on an unstructured list.

To answer the question of applicability we define the assignment of the elements of the requirements specification in Fig. 1 to the elements of the generic as well as the domain-specific requirements model. We consider a requirements model to be applicable for the running example if the assignment is reasonable and complete. To ensure a reasonable assignment we discuss the assignment until a consensus is found among the authors. This method is also called researcher triangulation. We consider an assignment to be complete if all requirements model elements are used at least once. By this constraint we aim to prevent the introduction of unnecessary or duplicate items to the requirements model.

To answer the question of efficiency we calculate the worst case effort for each modification approach by measuring the maximum number of requirements that have to be examined for determining impacted elements of the requirements specification. This means we count the candidates for an impact that need to be examined in the worst case after each modification operation (+ and Δ) for the different impact analysis approaches. By worst case we mean that during an impact analysis step we do not rely on knowledge obtained from previous examinations. We calculate the efficiency of the approaches by calculating the relative effort of the generic and the domain specific requirements model assignments to the unstructured requirements list without a requirements model.

5.2 Results

We first elaborate on the applicability of our approach for the running example before assessing its efficiency with respect to the other modification approaches.

(a) *Applicability* The results of mapping the elements of the requirements specification from Fig. 1 with the generic and the domain-specific requirements model are shown in

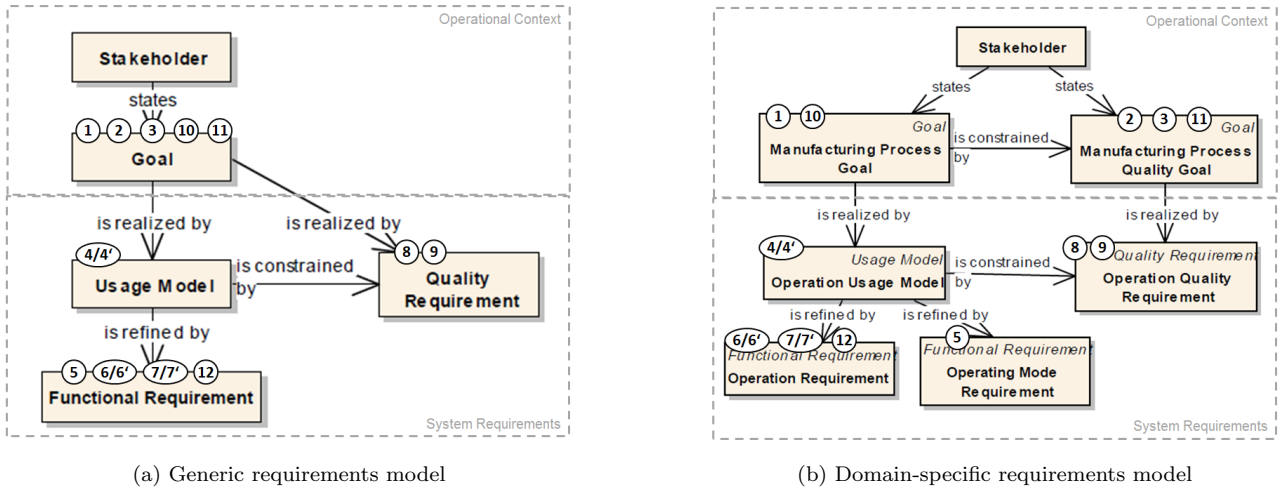


Fig. 6. Manual classification of the requirements (see Fig. 1) using the different requirements models (see Fig. 4).

Impact analysis approach	Effort in number of examined requirements						Absolute effort	Relative effort
	+ 10	+ 11	Δ 4'	Δ 6'	Δ 7'	+ 12		
(a) Without requirements model	9	10	10	10	10	11	60	100%
(b) With generic requirements model	6	7	5	2	2	3	25	42%
(c) With domain-specific requirements model	4	4	5	1	1	2	17	28%

Fig. 7. Modification effort in number of requirements examined per modification operation.

Figure 6. All requirements model elements are used at least once and thus we consider the model to be complete regarding the case study. In the case of the generic requirements model we obtain five goals, one scenario, four functional requirements, and two quality requirements. In the case of the domain-specific requirements model we obtain two manufacturing process goals, three manufacturing process quality goals, one operation scenario, one operation mode, three operation requirements, and two operation quality requirements. Elements that are modified in version 2 like the usage model are counted only once. Since the assignment to both the generic and domain-specific requirements models is reasonable by construction, we conclude that the requirements models are applicable for the running example.

(b) Efficiency The results of estimating the modification effort for the three different modification approaches are summarized in Fig. 7. The rows of the table show the three modification approaches (without requirements model, with generic requirements model and with domain-specific requirements model). The columns show the sequence of the modification operations (adding or changing requirements). Then, for each modification approach and modification operation the worst case effort is calculated in number of requirements that need to be examined after applying the modification. The last two columns summarize the absolute effort as well as the relative effort with respect to the maximum effort for each of the three modification approaches. The algorithm is applied to all modifications for all three approaches. For illustrative purposes we describe how the algorithm was applied to the first three modifications in each modification approach.

When working **(a) without a requirements model**, in total 60 requirements need to be examined in the worst

case for the six modification operations. After adding the label requirement 10, the existing nine requirements from version 1 need to be examined in order to determine that the operating scenario 4 needs to be changed. Subsequently, the plastic requirement 11 is added, leading to another examination of the other ten requirements to find out that the operating scenario 4 needs to be changed as well. After changing 4, the other ten requirements need to be examined in order to identify necessary changes in requirements 6 and 7.

For the impact analysis **(b) with the generic requirements model**, in total only 25 requirements need to be examined in the worst case, amounting to 42% of the original effort. After adding the label goal 10, the other three goals, the scenario and the two quality requirements need to be examined in order to determine the change necessary in the operating scenario 4. In this step, the three functional requirements can be omitted with respect to the unstructured list. Subsequently, the plastic goal 11 is added, leading to another examination of the other four goals, the usage model and the two quality requirements to identify the change in the operating scenario 4. Again the three functional requirements can be omitted. After changing 4, the three functional requirements and two quality requirements need to be examined to identify the change required in the functional requirements 6 and 7. The five goals and the usage model can be ignored.

Finally, when working **(c) with the domain-specific requirements model**, in total only 17 requirements need to be examined in the worst case, which amounts to 28% of the original effort and 68% of the effort when using the generic requirements model. After adding the label manufacturing process goal 10, the other manufacturing process goal, the two manufacturing process quality goals

and the operation usage model need to be examined in order to determine the change necessary in the operating scenario 4. Compared with the generic requirements model, the two operation quality requirements can be omitted. Subsequently, the plastic requirement 11 (manufacturing process quality goal) is added. This leads to another examination of the two manufacturing process quality goals, the operation usage model and the two operation quality goals to find out that scenario 4 needs to be changed. The two manufacturing process goals can be ignored compared with the generic requirements model. After changing 4, the operating mode requirement, two operation requirements and two operation quality requirements need to be examined to identify the changes required in operation requirements 6 and 7. In this step no extra efforts can be saved with respect to the generic requirements model.

In our example we showed that using a domain-specific requirements model during impact analysis is more efficient than using no requirements model or a generic requirements model. Given the case study and the worst case effort estimate, the domain-specific requirements model needs only 28% of the effort of without a requirements model and 68% of the effort of the generic requirements model. In particular, the effort reduction is explained by a more fine-grained distinction between process goals and process quality goals (as opposed to generic goals). Similarly, the distinction between operating mode requirements and operations requirements (as opposed to generic functional requirements) provides a benefit. Note that in this case the choice of the modification approach does not influence the quality of the impact analysis - in all three cases identical impacts in the requirements specification are determined.

5.3 Discussion

The study shows that the requirements models are both suitable for the running example and efficient for impact analysis in our case. However, the executed study includes some limitations, which are discussed in the following.

We have measured applicability by means of a reasonable assignment between the requirements and the requirements model elements such that every requirements model element is used at least once. The constraint that every content item is used at least once provides an objective measure. However, the question of reasonable assignments is inherently subjective. In the study we address this issue by deriving the mapping in a team of researchers both from the fields of computer science and automation systems engineering. While we gain some objectivity through this measure, researcher bias still influences the mapping. To achieve objectivity we would have to conduct an independent replication study including a range of industry partners, each carrying out the assignments independently.

We have estimated the worst case effort for each modification approach by counting the maximum number of requirements that have to be examined during the manual selection of actually impacted requirements. We assume that the number of examined requirements interrelates with the effort required for carrying out the modifications. The study assumes that the modifications are executed strictly sequentially and after each modification

operation the other requirements have to be examined for possible impacts. We compared the efficiency of the three approaches by calculating the relative effort of each approach to the unstructured list. Through this precise rule for calculating the efficiency we obtain objectivity in assessing the efficiency of our approach. However, the assessment provides a theoretical measure of efficiency, as the number of requirements could be reduced easily by performing several change impact analyses in iteration over the requirements. This holds as long as the various change impacts can be remembered by the analyst. In our opinion, the efficiency measure becomes more reasonable when the requirements specifications and the associated changes are more complex. This is because with added complexity it becomes ever harder to remember each individual change. We would therefore argue not only that our approach would scale, but that in fact the advantages would increase for larger specifications. We therefore believe that the theoretical efficiency measure would correlate well to empirical measures obtained in industry.

The comparison of the three modification approaches shows that the approach using a generic requirements model in theory can be more efficient than an unstructured list. An adaptation of the requirements model to the manufacturing automation systems domain might even lead to additional efficiency improvements. However, our theoretical measure of efficiency does not include a number of important factors that need to be considered in practice. For a concrete utilization of the approach, the effort of tailoring the requirements model with respect to actual project needs has to be considered. Furthermore, efforts like training the staff on the requirements model and manually classifying the requirements specification have to be incorporated, which both depend on the granularity of the model. Consequently, additional studies are required to understand the impact of these factors on the overall efficiency.

An open challenge remains the selection and tailoring of the requirements model. A systematic model selection would minimize the manual activities of the approach. To this end, we plan to refine the requirements model using relevant standards from the automation systems domain and evaluate its efficiency with input from industry experts. This provides further insights into the relation between requirements model granularity and efficiency. In order to increase the benefits of the approach, we plan to extend the approach so that impact analysis can be performed seamlessly at the interface of requirements and architecture specification.

6. CONCLUSION

This paper presented a model-based approach for the efficient identification of impacts of a change on the requirements specification. The approach is demonstrated on the example of a pick and place unit and validated by this example in terms of applicability and efficiency. In the given setting we could validate the applicability of the proposed approach by means of a reasonable assignment from requirements specification to the underlying requirements model. In the example the application of a rather generic (not domain-specific) requirements model reduces the effort of analyzing the impact of a change to 42% compared

to an unstructured list. A requirements model specialized to the manufacturing automation systems domain further reduces the effort to 28%. In our case the choice of the modification approach does not influence the quality of the impact analysis - in all three cases identical impacts in the requirements specification are determined.

We conclude that the approach can improve the efficiency of impact analysis in the requirements specification for the manufacturing automation systems domain. The efficiency can increase further when adding domain-specific properties to the underlying model. Due to our notion of worst-case efficiency we argue that the impact analysis approach is particularly applicable in the case that many changes are necessary in a requirements specification, or where the specification is particularly large.

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